

BPHO ROUND 0 · MARKSCHEME

BPhO Round 0 — 2025 paper

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*The one real Round 0 paper in existence. Reproduced from the published BPhO material for personal study.*

Quick key

Q1 E	Q2 C	Q3 C	Q4 B	Q5 C
Q6 B	Q7 C	Q8 E	Q9 A	Q10 A
Q11 A	Q12 A	Q13 D	Q14 D	Q15 B
Q16 B	Q17 E	Q18 A	Q19 C	Q20 D
Q21 A	Q22 A	Q23 B	Q24 A	Q25 E

Worked solutions

Q1 · Work done stretching a wire — **Answer E**

What is being tested. The factor of $\frac{1}{2}$ in stored energy — where it comes from, and whether you check units before choosing between two options that differ by nothing else.

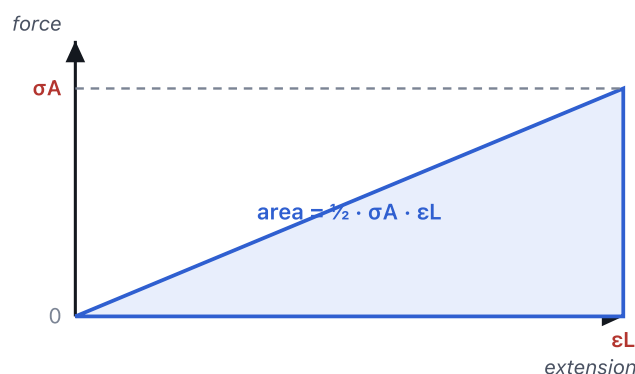
Step 1 — turn stress and strain into force and extension. The definitions do the work for you:

$$\begin{aligned}\sigma &= F / A \quad \Rightarrow \quad F = \sigma A && \text{(the force on the wire)} \\ \epsilon &= \Delta L / L \quad \Rightarrow \quad \Delta L = \epsilon L && \text{(how far it stretches)}\end{aligned}$$

Step 2 — the work done is the area under the force–extension graph. A wire obeying Hooke's law does not jump to its full load: the force rises from zero to σA exactly in step with the extension growing from zero to ϵL . The graph is therefore a triangle, and the work done is its area:

$$W = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times (\epsilon L) \times (\sigma A) = \frac{1}{2} LA\sigma\epsilon$$

Step 3 — the same answer from energy density, as a cross-check. The strain energy stored per unit volume is $\frac{1}{2}\sigma\epsilon$; multiply by the volume AL and you recover $\frac{1}{2}LA\sigma\epsilon$. Two independent routes, one answer.



The force rises in step with the extension. Because the wire behaves elastically, the force grows linearly from zero to σA as the extension grows from zero to ϵL , so the work done is the shaded triangle — not the rectangle you would get if the full force were applied throughout.

Answer: E, $\frac{1}{2}LA\sigma\epsilon$.

Why the other four are wrong. Option A is the same product with the $\frac{1}{2}$ thrown away: $LA\sigma\epsilon$ is the work you would do if the wire took its full load from the first instant and never resisted gradually — which is precisely what an elastic body does not do. The other three are not energies at all. B, $\sigma A / (\epsilon L)$, has the units of N m^{-1} — a stiffness. C, $\sigma L / (2\epsilon A)$, has N m^{-3} . D, $\epsilon A / (\sigma L)$, has $\text{m}^3 \text{N}^{-1}$. None of those is a joule.

Do it the fast way. You never need the $\frac{1}{2}$ versus no- $\frac{1}{2}$ decision to be a coin toss. Check the limiting case $\epsilon \rightarrow 0$: with no stretch, no work has been done, so the answer must vanish. B does the opposite — it diverges — and dies immediately. Then A and E are the only survivors, and the triangle versus rectangle argument settles them.

The wider point. The $\frac{1}{2}$ keeps appearing wherever two quantities grow together from zero: $\frac{1}{2}kx^2$, $\frac{1}{2}mv^2$, $\frac{1}{2}QV$, $\frac{1}{2}CV^2$, and here $\frac{1}{2}\sigma\epsilon$ per unit volume. Any stored-energy expression with no $\frac{1}{2}$ still on the table deserves a second look.

Relevant topics: strain energy; stress and strain; dimensional checking.

The trap. Picking $LA\sigma\epsilon$ and dropping the $\frac{1}{2}$. The energy stored in a gradually loaded elastic wire is half the final force times the final extension.

Q2 · Snell's law with angles measured from the surface — Answer C

The whole question is in the diagram. In this figure the two angles are drawn between each ray and the **surface**, not between each ray and the normal. Every textbook writes Snell's law in the normal form, and quoting it from memory is what makes this question catch people: option A is the formula you have written a hundred times, applied to angles that are not the ones it belongs to.

Convert, then use Snell. Let i and r be the angles to the normal. Since the normal is perpendicular to the surface,

$$\begin{aligned} i &= 90^\circ - \theta_1 & r &= 90^\circ - \theta_2 \\ n_1 \sin i &= n_2 \sin r \\ n_1 \sin(90^\circ - \theta_1) &= n_2 \sin(90^\circ - \theta_2) \\ n_1 \cos \theta_1 &= n_2 \cos \theta_2 \end{aligned}$$

Answer: C, $n_1 \cos \theta_1 = n_2 \cos \theta_2$.

Check it against a case you know. The figure is drawn for $n_1 = 1.5$ and $n_2 = 1$ with $\theta_1 = 60^\circ$. Then $\cos \theta_2 = 1.5 \times \cos 60^\circ = 0.75$ — and **stop there**. You do not need θ_2 itself: “41.4°” is a number no one can produce without a calculator, and nothing in this question asks for it. What you need is only whether θ_2 is larger or smaller than θ_1 , which $\cos \theta_2 = 0.75 > \cos 60^\circ = 0.5$ answers immediately: $\cos \theta_2$ is bigger, so θ_2 is the smaller angle, so the ray bent away from the normal. (Snell in the normal form says the same thing: $1.5 \sin 30^\circ = 0.75 = 1 \times \sin \theta_2'$ where $\theta_2' = 48.6^\circ$ from the normal — again a number you never have to write down.) The ray leaves the glass at a smaller angle to the surface, bending away from the normal — exactly as light does on the way out of a denser medium.

The limiting case that kills A. Slide the incident ray down until it travels along the interface: $\theta_1 \rightarrow 90^\circ$. In this picture that means the ray is travelling parallel to the boundary, and it should keep travelling parallel to it — nothing in the problem singles out either side. Option C gives $\cos \theta_1 \rightarrow 0$, hence $\cos \theta_2 \rightarrow 0$, hence $\theta_2 \rightarrow 90^\circ$: still along the boundary. ✓ Option A would give $\sin \theta_2 = n_1/n_2$, a definite angle to the surface, so the same limiting ray would suddenly refract — nonsense.

Where the other options come from. A is Snell's law with the angles to the normal — the wrong angles for this diagram. B is A with the refractive indices swapped. D is C with the indices swapped. E mixes a sine with a cosine, which no physical law does.

The transferable rule. Before substituting into any standard formula, read which angle the diagram actually marks. A marked angle is a fact about the figure; a remembered formula is a fact about a different figure until you have checked the two agree.

Relevant topics: refraction; Snell's law; refractive index; the small-angle and limiting-case checks.

The trap. Answering with the remembered $n_1 \sin \theta_1 = n_2 \sin \theta_2$. The diagram measures both angles from the surface, so the equation is $n_1 \cos \theta_1 = n_2 \cos \theta_2$.

Q3 · Which of these is not a unit of impulse — **Answer C**

Read the negative. Three of the five are genuine units of impulse and you are asked for the odd one out. Note that this is a “which is not” question: a slip here costs a mark you had already earned.

Fix the target. Impulse is force $\times$ time, and it equals the change in momentum, so its dimensions are those of momentum:

$$[\text{impulse}] = \text{N s} = \text{kg m s}^{-1}$$

Reduce all five to base units. Use $\text{N} = \text{kg m s}^{-2}$, $\text{W} = \text{J s}^{-1} = \text{kg m}^2 \text{s}^{-3}$, $\text{Pa} = \text{N m}^{-2} = \text{kg m}^{-1} \text{s}^{-2}$, $\text{C V} = \text{J} = \text{kg m}^2 \text{s}^{-2}$.

A	N s	$= \text{kg m s}^{-2} \cdot \text{s}$	$= \text{kg m s}^{-1}$	✓
B	$\text{W s}^2 \text{ m}^{-1}$	$= \text{kg m}^2 \text{s}^{-3} \cdot \text{s}^2 \cdot \text{m}^{-1}$	$= \text{kg m s}^{-1}$	✓
C	Pa m^3	$= \text{kg m}^{-1} \text{s}^{-2} \cdot \text{m}^3$	$= \text{kg m}^2 \text{s}^{-2}$	✗ (= J)
D	kg m s^{-1}		$= \text{kg m s}^{-1}$	✓
E	C V s m^{-1}	$= \text{kg m}^2 \text{s}^{-2} \cdot \text{s} \cdot \text{m}^{-1}$	$= \text{kg m s}^{-1}$	✓

Answer: C, Pa m^3 — which reduces to the joule. Pascal cubic metres is a unit of energy (it is the work done by a pressure pushing out a volume), not of impulse.

Why B and E look wrong but are not. They are the ones that make this question worth setting. B is a watt second squared per metre: the awkwardness is only in the presentation, not in the physics. E is a coulomb volt second per metre, and since a coulomb volt *is* a joule there is nothing exotic in it either. Both collapse to kg m s^{-1} . There are infinitely many combinations of units that do.

The technique that settles it in fifteen seconds. Do not judge a unit by how familiar it looks. Convert it. Judge it only after it is in base units, and if a candidate contains a joule (or anything that reduces to one) you know immediately that it is energy, not momentum.

Relevant topics: base and derived units; impulse and momentum; dimensional consistency.

The trap. Rejecting $\text{W s}^2 \text{ m}^{-1}$ or C V s m^{-1} because they do not look like N s . Both reduce to kg m s^{-1} ; it is Pa m^3 that reduces to an energy.

Q4 · Alpha and beta accounting in a decay series — **Answer B**

What is being tested. That an alpha decay and a beta decay change the two nuclear numbers differently — and that a decay series has to balance both of them.

The two bookkeeping rules. An alpha particle is a helium nucleus, so

alpha (α): $A \rightarrow A - 4$, $Z \rightarrow Z - 2$
 beta minus (β^-): $A \rightarrow A$, $Z \rightarrow Z + 1$

Step 1 — count the alphas from the mass number. Only alpha decay changes A , and each one removes 4:

$\Delta A = 238 - 206 = 32$
 number of alphas = $32 / 4 = 8$

Step 2 — see where those eight alphas leave the charge. Eight alphas remove $8 \times 2 = 16$ units of charge:

Z after the alphas = $92 - 16 = 76$

Step 3 — close the gap with beta minus decays. Lead-206 has $Z = 82$, so the series must gain $82 - 76 = 6$ units of charge, and a beta minus decay is the only step here that adds one:

number of beta minus decays = $82 - 76 = 6$

Answer: B, 6.

Check the whole equation. Eight alphas and six beta-minuses:

A: $238 - 8(4) = 206$ ✓
 Z: $92 - 8(2) + 6(1) = 82$ ✓

Both nuclear numbers balance, so the count is right.

Where the wrong options come from. Option C, 8, is the number of alphas — the answer to a question that was not asked, and the commonest slip of all: the two numbers feel interchangeable but only one of them is 8. Options D, 10, and E, 12, come from arithmetic on the charge gap with the wrong particle:

treating a beta decay as though it changed the charge by 2 (the alpha's step) gives $16/2 + \dots$ type answers, and A, 4, is half the correct count, as if a beta decay carried two units of charge.

The wider point. Every nuclear reaction balances A and Z separately. Do the two sums independently — mass number first, because only the alpha moves it, then charge — and then verify by adding both columns back up.

Relevant topics: alpha and beta decay; conservation laws in nuclear decay; decay series.

The trap. Reporting 8, the number of alpha decays. The question asks for the beta minus decays, which the charge balance alone fixes at 6.

Q5 · Interference at a given path difference — Answer C

Turn the path difference into a number of wavelengths. That single step decides everything:

$$\Delta x / \lambda = 7.5 / 1.8 = 75/18 = 25/6 = 4 + 1/6 \text{ wavelengths}$$

Do that division as a fraction, not on a calculator. $7.5/1.8 = 75/18$; both divide by 3, so it is $25/6$, and $25/6 = 4$ remainder 1, so **4 whole wavelengths plus $1/6$** . The “4” is what misleads people who reach for a decimal and see 4.1667. Keeping the fraction makes the remainder obvious — and the remainder is the only part that matters. Four whole wavelengths contribute nothing — a shift of a whole wavelength is a shift of a whole cycle. What is left over is the fractional part:

$$\begin{aligned} \text{fractional part} &= 1/6 \text{ of a wavelength} \\ \text{phase difference} &= 2\pi \times 1/6 = 60^\circ \end{aligned}$$

Six degrees of a cycle is a long way from half a cycle. At 60° the two waves are much more in step than out of step. The resultant amplitude is

$$2A \cos(\phi/2) = 2A \cos 30^\circ = 1.73A$$

against $2A$ for perfect reinforcement — 87% of it. That is “mostly constructive”.

Answer: C, mostly constructive.

Why not fully constructive. That requires the path difference to be a whole number of wavelengths, and $25/6$ is not a whole number. The test that makes this quick: compare the remainder $1/6$ against the midpoint $1/4$. Since $1/6 < 1/4$ (cross-multiply: $4 < 6$), the shift is closer to “in step” than to “out of step” — so the answer is “mostly”, on the constructive side. No decimal needed anywhere.

Why not E, not enough information. The two waves are coherent, they come from the same source, and they arrive with equal amplitude. Given the wavelength and the path difference, the phase relationship at the detector is completely determined. What is *not* determined is the absolute amplitude — you cannot say the detector reads $1.73A$ without knowing A — but the question asks about the type of interference, which is a question about phase, and phase is fixed.

The general rule worth carrying into the exam. Work in fractions of a wavelength: a whole number gives fully constructive, a half-odd number ($n + \frac{1}{2}$) gives fully destructive, and anything else is “mostly” whichever side of the midpoint it falls. Then convert to radians ($\times 2\pi$) or degrees ($\times 360^\circ$) only if the question asks for them.

Relevant topics: superposition; path and phase difference; coherence; interference conditions.

The trap. Turning 7.5/1.8 into 4.1667 on a calculator, rounding it to 4, and calling it fully constructive — or treating “from the same source” as a reason to doubt the information given.

Q6 · Cheapest network of exactly $4\ \Omega$ — **Answer B**

What is being tested. Whether you can search a small space of possibilities systematically. The physics — series and parallel — is trivial; the mark is for being organised under time pressure.

Step 1 — find the obvious solution so you know the price you have to beat. The parallel combination that everyone spots is

$$6\ \Omega \parallel 12\ \Omega = (6 \times 12)/(6 + 12) = 4\ \Omega, \quad \text{cost } £3 + £2 = £5$$

So the answer is at most £5. Options D (£6) and E (£7) can be ignored from here on, and the remaining question is whether £3 or £4 can do it.

Step 2 — ask what £3 can buy, exhaustively. With £3 you can buy any of these (there is no change to save, so nothing else fits):

one $2\ \Omega$ (£3) or one $6\ \Omega$ (£3), or three $3\ \Omega$ (£3) or one $12\ \Omega$ + one $3\ \Omega$ (£3), plus any network you like built only from $3\ \Omega$ resistors

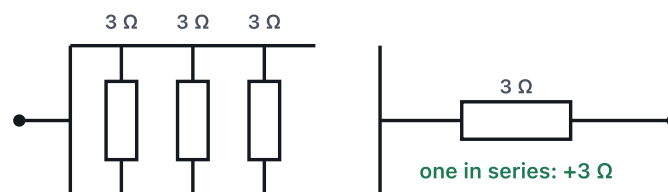
Now list what each of those can actually make:

one $2\ \Omega$	→ $2\ \Omega$	
one $6\ \Omega$	→ $6\ \Omega$	
one $12\ \Omega$	→ $12\ \Omega$	
one $12\ \Omega$ and one $3\ \Omega$	→ $15\ \Omega$	or $2.4\ \Omega$
one $3\ \Omega$	→ $3\ \Omega$	
two $3\ \Omega$	→ $6\ \Omega$	or $1.5\ \Omega$
three $3\ \Omega$ (any arrangement)	→ $9\ \Omega, 4.5\ \Omega, 2\ \Omega, 1\ \Omega$	

Not one of those is $4\ \Omega$, and nesting the network differently cannot help: with n identical equal resistors the achievable values are a short, fixed list, and a lone $2\ \Omega$, $6\ \Omega$ or $12\ \Omega$ cannot be turned into $4\ \Omega$ by itself. **So £3 is impossible.**

Step 3 — hit exactly £4. Buy four $3\ \Omega$ resistors for £4 and put three of them in parallel:

three $3\ \Omega$ in parallel: $3 / 3 = 1\ \Omega$
 in series with the fourth: $1\ \Omega + 3\ \Omega = 4\ \Omega, \quad \text{cost } 4 \times £1 = £4$



three in parallel: $3/3 = 1\ \Omega$, cost £3

$$R = 1\ \Omega + 3\ \Omega = 4\ \Omega \text{ for } \pounds 4$$

The winning combination. Three $3\ \Omega$ resistors side by side give $3/3 = 1\ \Omega$; one more $3\ \Omega$ in series brings it to exactly $4\ \Omega$. Three of them cost £1 each, the fourth another £1.

There is a second £4 solution, worth seeing because it shows the same price from a different direction: buy one $12\ \Omega$ and two $3\ \Omega$, put the $3\ \Omega$ pair in series and that across the $12\ \Omega$ —

$$12\ \Omega \parallel (3\ \Omega + 3\ \Omega) = 12\ \Omega \parallel 6\ \Omega = 4\ \Omega, \quad \text{cost } \pounds 2 + \pounds 1 + \pounds 1 = \pounds 4$$

Answer: B, £4.

Why the other options are wrong. A, £3, is impossible — the complete list above shows it. C, £5, is the answer for anyone who stops at the first combination they think of ($6\ \Omega \parallel 12\ \Omega$); it is a real solution, just not the cheapest. D, £6, is what you get by building $4\ \Omega$ out of two $2\ \Omega$ resistors in series, or three $12\ \Omega$ in parallel: correct, and twice the price. E, £7, is $3\ \Omega + (2\ \Omega \parallel 2\ \Omega)$, or many other dearer versions of the same idea.

The habit that makes this safe. Price first, physics second. Establish an upper bound by finding any solution, then work upward from the cheapest purchase that could possibly do the job, and eliminate. One cheap combination that must be beaten plus one exhaustive list of what the cheaper budget can buy is a complete proof — and it takes less time than hoping a cleverer network will occur to you.

Relevant topics: series and parallel resistors; equivalent resistance; systematic search under time pressure.

The trap. Settling for £5 because $6\ \Omega \parallel 12\ \Omega = 4\ \Omega$ springs to mind. Four $3\ \Omega$ resistors do it for £4 — three in parallel make $1\ \Omega$, and one in series adds the other $3\ \Omega$.

Q7 · Relative velocity: catching up — Answer C

Read the arrows before you do any arithmetic. Both velocity arrows point the same way: A moves at $6\ \text{m s}^{-1}$ with B ahead of it at $3\ \text{m s}^{-1}$. Same direction, so the gap closes only at the difference:

$$\text{closing speed} = 6 - 3 = 3\ \text{m s}^{-1}$$

Step 1 — time to close 18 m at $3\ \text{m s}^{-1}$.

$$t = 18 / 3 = 6\ \text{s}$$

Step 2 — how far A travels in that time. A moves at its own $6\ \text{m s}^{-1}$, not at the closing speed:

$$s_A = 6 \times 6 = 36 \text{ m}$$

Answer: C, 36 m.

Check it in one line. In 6 s, B travels $3 \times 6 = 18 \text{ m}$. A started 18 m behind, so it must travel B's 18 m plus the 18 m gap it has to make up:

$$s_A = 18 + 18 = 36 \text{ m} \quad \checkmark$$

Both routes agree, which is the point of doing the check: the second one catches an error in the first if you have mixed up which speed belongs to whom.

The trap the paper is hunting. If the two particles were approaching each other, the closing speed would be $6 + 3 = 9 \text{ m s}^{-1}$, the time would be $18/9 = 2 \text{ s}$, and A would travel 12 m. **12 m is not among the options.** That is deliberate: it is a signal, not an accident. If your working leads to 12 m, you have read the diagram as two particles approaching, and you should go back and look at the arrowheads rather than look for your answer in the list.

Where the other options come from. A, 18 m, is the distance B travels in the same 6 s (or A's own share of the initial separation). D, 45 m, and E, 54 m, are the results of using the wrong speed for A: 54 m is $9 \text{ m s}^{-1} \times 6 \text{ s}$ — the closing speed of a head-on approach carried through the correct same-direction time — and B, 27 m, is the mean of the two speeds (4.5 m s^{-1}) sustained for the full 6 s.

The general method for a catch-up question. Set up $v_A t = v_B t + \text{gap}$ and solve once. For this paper that is $6t = 3t + 18$, so $3t = 18$, $t = 6 \text{ s}$, then substitute back for whichever distance is asked. One equation, no simultaneous reasoning, and it never confuses the two speeds.

Relevant topics: relative velocity; constant-velocity kinematics; reading a velocity diagram.

The trap. Adding the speeds and getting 12 m — which is not an option. The arrows show the particles travelling the same way, so the closing speed is $6 - 3 = 3 \text{ m s}^{-1}$.

Q8 · Ideal voltmeter and ammeter readings — **Answer E**

What is being tested. Whether you know what “ideal” means for each meter — and whether you can then see that two of the five drawn components are doing nothing at all.

Step 1 — let the meters simplify the circuit for you.

ideal ammeter: resistance 0 $\Rightarrow$ the branch it sits in is a short circuit
 ideal voltmeter: resistance ∞ $\Rightarrow$ the branch it sits in carries no current

The ammeter is in parallel with the bottom resistor, so that resistor has zero potential difference across it and carries no current. The voltmeter is in series in the bottom-left lead, so that lead carries no current either — and since it carries none, the junction it connects to is not fed from there. What survives is a single loop:

cell $\epsilon \rightarrow$ top resistor $R \rightarrow$ right-hand node
 right-hand node $\rightarrow$ (ammeter, 0Ω) $\rightarrow$ middle node $\rightarrow$ middle resistor $R \rightarrow$ left-hand node
 $\rightarrow$ back to the cell

Step 2 — the loop. Two resistors of R in series with the cell, so the total resistance is $2R$ and the current is

$$I = \varepsilon / (2R)$$

Every element in that loop carries this same current, the ammeter included. So:

$$\text{ammeter reading} = I = \varepsilon / (2R)$$

Step 3 — the voltmeter. It is connected across the middle resistor, whose resistance is R and whose current is I :

$$V = I R = (\varepsilon / 2R) \times R = \varepsilon / 2$$

Answer: E — voltmeter $\varepsilon/2$, ammeter $\varepsilon/(2R)$.

Cross-check by potential. The cell lifts the potential by ε ; the two equal resistors in the loop divide it equally, so each takes $\varepsilon/2$. The voltmeter spans one of them and reads $\varepsilon/2$. The bottom resistor, with no current, takes no share of the potential difference at all — which is exactly why it can be ignored.

Why the other options fail. A, B and C all quote a current of $\varepsilon/(2R)$ or ε/R paired with a voltmeter reading of $\varepsilon/3$. That $\varepsilon/3$ comes from treating all three resistors as carrying the loop current and sharing the emf in three equal parts — which cannot happen, because the ammeter's zero resistance takes the bottom resistor out of the circuit. C and D pair a current with a voltage that belong to different assumptions about the network, and A assumes a total resistance of $3R$, i.e. the naive “three resistors in series”.

The habit to take away. In any question with a meter drawn on it, ask two questions before you do anything else: which branch has *zero* resistance (that resistor is shorted out), and which branch has *infinite* resistance (that resistor carries nothing). Most “impossible” circuit questions in Round 0 are two-component questions wearing three resistors.

Relevant topics: ideal meters; series and parallel networks; emf and potential difference; node reasoning.

The trap. Using all three resistors and reading a total resistance of $3R$. The ideal ammeter short-circuits the resistor beside it, so the cell drives current through two resistors, not three.

Q9 · First-harmonic frequency against resistance — Answer A

What is being tested. A two-step chain of proportionality — waves, through density, to electrical resistance — where every shared constant cancels, so you never need a value.

Step 1 — write the frequency in terms of the wire's geometry. The first harmonic on a string fixed at both ends has half a wavelength along the wire, so

$$f = v / (2L), \quad v = \sqrt{T/\mu}, \quad \mu = \rho A$$

$$\Rightarrow f = (1 / 2L) \sqrt{T / (\rho A)}$$

Step 2 — keep only what differs. The two wires share the length L , the tension T , the material (so ρ is the same) and the working frequency formula. Everything cancels except the cross-section:

$$f \propto 1 / \sqrt{A} \quad \Rightarrow \quad A \propto 1 / f^2$$

Step 3 — the frequency ratio becomes an area ratio. With $f_1 : f_2 = 2 : 1$:

$$A_1 / A_2 = (1/f_1^2) / (1/f_2^2) = (f_2 / f_1)^2 = (1/2)^2 = 1/4$$

So wire 2 has four times the cross-sectional area of wire 1. The wire that sounds the higher note is the thinner one.

Step 4 — convert area into electrical resistance. For a uniform wire of resistivity ρ_e ,

$$R = \rho_e L / A \quad \Rightarrow \quad R \propto 1 / A \quad (\text{same } L, \text{ same material})$$

$$R_1 / R_2 = A_2 / A_1 = 4$$

Answer: A, 4 : 1.

The physical reading. Doubling the frequency means quartering the area, and a quarter of the area is four times the resistance. Thin, high-pitched, high-resistance; thick, low-pitched, low-resistance. If your answer has the thicker wire with the higher frequency, or the ratio the wrong way up, this sentence tells you so without any algebra.

Why the other options are wrong — and a one-line argument that kills them all. The chain above gives $R \propto f^2$, and the frequency ratio is 2, so the resistance ratio must be $2^2 = 4$. Only option A is 4 (or 1/4). D, 2 : 1, is what you get by forgetting the square in $A \propto 1/f^2$ — that is, by treating the wave speed as independent of the cross-section, which would make the frequencies depend on nothing at all. E, $\sqrt{2} : 1$, comes from applying the square root at the wrong stage of the chain; B and C are combinations that cannot arise from a chain of powers of the ratio 2.

The general method. When a question gives you no numbers, write the quantity you want as a product of powers of the given quantities ($R \propto f^n$), and read the exponent off the algebra. Then the ratio is $(\text{ratio})^n$. Nothing needs a calculator, and a slip in the exponents shows up immediately as a ratio that is not a power of the given one.

Relevant topics: standing waves and the first harmonic; wave speed on a wire; mass per unit length; resistivity; ratio reasoning.

The trap. Getting the ratio the wrong way up, or forgetting the square. The higher-pitched wire is thinner ($f \propto 1/\sqrt{A}$) and thinner means more resistance, so $R \propto f^2$ and the answer is 4 : 1.

Q10 · Estimating the number of atoms in the Earth — **Answer A**

What is being tested. Whether you can turn a fact you know (the mass of the Earth) plus a fact you can make (the mass of a typical atom) into an order of magnitude — and whether you resist the urge to look for a calculator you are not allowed to have.

Step 1 — the mass of the Earth. This is worth knowing as an anchor: $M_\oplus \approx 6 \times 10^{24} \text{ kg}$.

Step 2 — the mass of a typical atom. The Earth is mostly iron, oxygen, silicon and magnesium, so a typical nucleus has a mass number of a few tens; take $A \approx 30$. One atomic mass unit is about $1.7 \times$

10^{-27} kg, so

$$m_{\text{atom}} \approx 30 \times 1.7 \times 10^{-27} \text{ kg} \approx 5 \times 10^{-26} \text{ kg}$$

Step 3 — divide.

$$\begin{aligned} N &= M / m_{\text{atom}} = (6 \times 10^{24}) / (5 \times 10^{-26}) \\ &= (6/5) \times 10^{50} \\ &\approx 1.2 \times 10^{50} \end{aligned}$$

Answer: A, about 10^{50} atoms.

The same estimate through moles, which is faster if you are comfortable with Avogadro. A mole of typical rock weighs about 30 g, so 1 kg of rock contains roughly $6 \times 10^{23} / 0.03 \approx 2 \times 10^{25}$ atoms. Multiply by the mass of the Earth:

$$N \approx (6 \times 10^{24} \text{ kg}) \times (2 \times 10^{25} \text{ atoms kg}^{-1}) \approx 1.2 \times 10^{50}$$

Carry this anchor: a kilogram of ordinary rock contains of order 10^{25} atoms. It converts a hard question into one multiplication.

Why the answer does not depend on the composition. Suppose the Earth were pure iron, $A = 56$: then $N \approx 6.4 \times 10^{49}$. Suppose it were pure oxygen, $A = 16$: $N \approx 2.2 \times 10^{50}$. Both are 10^{50} to the nearest power of ten. The options differ by factors of 10^5 , so an estimate good to within a factor of a hundred is easily enough — you are choosing between 10^{50} and 10^{55} , not between 1.2 and 1.3.

Where the other options come from. 10^{55} and beyond require an error of several orders of magnitude — the usual source is mishandling the atomic mass. Using the mass of a single nucleon (1.7×10^{-27} kg) instead of an atom still lands you at 4×10^{51} , which is still nearest to 10^{50} : that is the point about robustness. Only a slip in the exponents — writing the atomic mass as 10^{-22} kg, say, or losing the 10^{23} in Avogadro and treating it as 10^{20} — gets you to 10^{55} or beyond.

The technique. In an estimation question, collect the powers of ten first and the leading digits second. Do not tidy up. $(6 \times 10^{24}) / (5 \times 10^{-26})$ is $(6/5) \times 10^{50}$ and that is already an answer to the precision the options demand.

Relevant topics: estimation; Avogadro and the mole; standard form; rounding to an order of magnitude.

The trap. Reaching for a calculator — which is not permitted. The options are five powers of ten apart, so all the question wants is the exponent, and that comes from $(6 \times 10^{24}) / (5 \times 10^{-26})$.

Q11 · Counting the possible photon energies — Answer A

What is being tested. Whether you can see that this is a counting question in disguise. There is no physics beyond $hf = E_1 - E_2$; the mark is for setting the count up correctly and not double-counting.

Step 1 — one photon per pair of levels. The atom emits a photon when it drops from a higher level to a lower one, and the photon energy is fixed by which two levels are involved:

$$hf = E_{\text{upper}} - E_{\text{lower}}$$

So every pair of distinct levels contributes one possible photon energy. A pair, not an ordered pair: the transition from $n = 5$ down to $n = 2$ and the transition from $n = 2$ up to $n = 5$ describe the same energy gap, and only the downward one emits.

Step 2 — count the pairs. With ten levels there are

$$C(10, 2) = 10 \times 9 / 2 = 45$$

ways of choosing two distinct levels.

Answer: A, 45.

Where the word “maximum” comes from. In a real atom two different pairs can happen to have the same energy gap, and then two transitions produce identical photons and the count of *unique* energies is smaller. The question says “maximum”, which tells you to assume no such coincidences: 45 is the largest the count can be, achieved when all 45 gaps are different.

Where the wrong options come from — and how to catch them. All four of the others are products that count something slightly different:

$$\begin{array}{ll} 100 = 10 \times 10 & \text{ordered pairs including a level with itself (no transition)} \\ 90 = 10 \times 9 & \text{ordered pairs — each gap counted twice, once each way} \\ 110 = 11 \times 10 & \text{ordered pairs across eleven levels} \\ 55 = 11 \times 10 / 2 & \text{the unbiased pair count, but with eleven levels instead of ten} \end{array}$$

Answer: A, 45.

The check that takes two seconds. Whatever method you use, try it on three levels. Three levels give three pairs — $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$ — so the formula must produce 3 when $n = 3$. $n(n-1)/2 = 3$ ✓; $n^2 = 9$ ✗; $n(n-1) = 6$ ✗. Counting formulae are easy to verify on a tiny case and almost impossible to check by staring at them.

Relevant topics: energy levels and photon energies; the transition rule; combinations and counting.

The trap. Counting ordered pairs (90) or pairs including a level with itself (100). A photon energy belongs to an unordered pair of distinct levels: $C(10,2) = 45$.

Q12 · How the energy of an explosion is shared — **Answer A**

What is being tested. The consequence of momentum conservation in an explosion — and the discipline to test an algebraic answer rather than trust it.

Step 1 — the momenta are equal and opposite. The body started at rest, so the total momentum is zero before and after. Call the speed of the smaller fragment v_1 and of the larger v_2 :

$$m v_1 = (am) v_2 \quad \Rightarrow \quad v_1 = a v_2$$

The lighter fragment is the faster one, by exactly the mass ratio.

Step 2 — write both kinetic energies through the shared momentum. This is the step that makes the algebra clean. Both fragments carry the same magnitude of momentum, p , so use $KE = p^2/2m$:

$$\begin{aligned} \text{KE}(\text{small}) &= p^2 / (2m) \\ \text{KE}(\text{large}) &= p^2 / (2am) \end{aligned}$$

Same p , twice the mass, so the larger fragment takes the *smaller* share of the energy.

Step 3 — total, then ratio.

$$E = p^2/(2m) + p^2/(2am) = (p^2/2m)(1 + 1/a) = (p^2/2m) \cdot (a + 1)/a$$

Divide the kinetic energy of the larger mass by the total:

$$\begin{aligned} \text{KE}(\text{large}) / E &= [p^2/(2am)] / [(p^2/2m)(a+1)/a] \\ &= [1/(am)] \times [am/(a+1)] \\ &= 1/(a + 1) \end{aligned}$$

Answer: A, $\text{KE}(\text{larger}) = E/(1 + a)$.

Test it the way you should test every algebraic answer — on the ends.

- $a \rightarrow 1$: the masses are equal, so the energy splits equally.
 $E/(1 + a) \rightarrow E/2$ ✓
- $a \rightarrow \infty$: the big fragment is infinitely heavy, so it barely moves and takes almost no kinetic energy. $E/(1 + a) \rightarrow 0$ ✓

Both limits are right, and they already kill two options outright. Option C gives E as $a \rightarrow \infty$ — all the energy in the heaviest fragment, which is exactly backwards. Option D does the same. Option E, E/a , gives E at $a = 1$, where equal masses must split the energy equally: dead.

One numerical case to separate the last two. The $a \rightarrow 1$ limit does not distinguish A from B, because $a/(1 + a^2) \rightarrow 1/2$ as well. Put $a = 2$: masses m and $2m$, so $v_1 = 2v_2$, and the kinetic energies go as $m(2v)^2 = 4mv^2$ for the light one and $2m v^2$ for the heavy one — a 2:1 split, with the heavy fragment taking one third:

$$\begin{aligned} \text{A: } E/(1 + 2) &= E/3 = 0.333E & \checkmark \\ \text{B: } 2E/(1 + 4) &= 0.400E & \times \end{aligned}$$

Answer: A.

The physical reading worth keeping. Momentum is shared; energy is not. Two fragments of unequal mass fly apart with equal and opposite momenta, and because $\text{KE} = p^2/2m$, the *lighter* fragment carries most of the kinetic energy. At $a = 2$ the light fragment gets two thirds. Option B, $aE/(1+a^2)$, is the value you get by assuming the energy is shared in proportion to mass — the misconception the question is built to detect.

Relevant topics: momentum conservation in an explosion; kinetic energy in terms of momentum; limiting-case testing.

The trap. Assuming the heavier fragment takes the larger share of the energy. Equal momenta and $\text{KE} = p^2/2m$ mean the heavier one takes $E/(1+a)$ — less than half.

What is being tested. Pure dimensional analysis — “dependent only upon” is the instruction to ignore the physics and match dimensions. AQA does not examine this, so it is cheap BPhO-specific marks.

Step 1 — write the dimensions of everything involved.

pressure	$[p] = \text{N m}^{-2} = \text{kg m}^{-1} \text{s}^{-2} = \text{M L}^{-1} \text{T}^{-2}$
Planck	$[h] = \text{J s} = \text{kg m}^2 \text{s}^{-1} = \text{M L}^2 \text{T}^{-1}$
light	$[c] = \text{m s}^{-1} = \text{L T}^{-1}$
separation	$[r] = \text{m} = \text{L}$

Step 2 — assume a product form and match each base dimension. Put $p = k h^\alpha c^\beta r^\gamma$ with k a dimensionless constant:

M:	$\alpha = 1$
L:	$2\alpha + \beta + \gamma = -1$
T:	$-\alpha - \beta = -2$

Step 3 — solve the columns in order.

from M:	$\alpha = 1$
from T:	$\beta = 2 - \alpha = 1$
from L:	$\gamma = -1 - 2\alpha - \beta = -1 - 2 - 1 = -4$

Answer: D, $p \propto 1/r^4$. The full form is $p = k h c / r^4$, and the physics adds nothing to the exponent: the real result is $p = \pi^2 \hbar c / (240 r^4)$.

Why the tidy answer is the right one. If you have met the Casimir effect you may know that the force falls off as $1/r^4$; dimensional analysis has just produced it without any quantum electrodynamics. That is the point of the technique, and it is why the question can be set at all in a non-calculator paper.

Where the neighbouring options come from. Every wrong option is an offspring of one sign or one term in the L equation. Writing the length dimension of pressure as $+1$ instead of -1 — a pressure is a force per area, so the area's m^{-2} is easy to drop — gives $2 + 1 + \gamma = +1$, so $\gamma = -2$: option B. Losing the 2α contribution of h gives $\beta + \gamma = -1$, so $\gamma = -2$ again. Adding instead of subtracting the hc contributions gives $\gamma = -5$: option E. Option C is the value you get with a single exponent error, and A is essentially unforced.

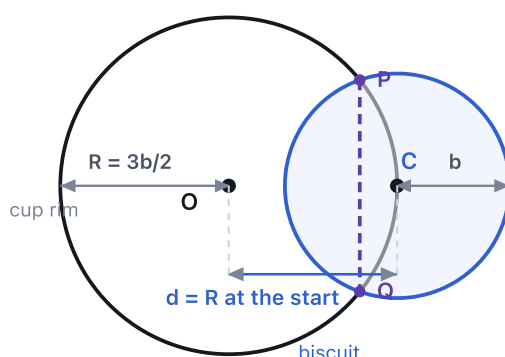
The technique, stated once for the whole paper. Three base dimensions means three equations, which means you can solve for three unknown exponents. Write the four rows out in full — pressure, h , c , r — solve the mass row first because it is always the shortest, then time, then length. Do not try to do it in your head: the question is worth one mark and the tableau takes twenty seconds.

Relevant topics: dimensional analysis; the dimensions of h and c ; the Casimir effect as a quantum pressure.

The trap. Getting one exponent wrong by writing the length dimension of pressure as positive. Force per area means $\text{M L}^{-1} \text{T}^{-2}$, and that L^{-1} is what makes the answer $1/r^4$ rather than $1/r^2$.

What is being tested. That “falling over” is a statement about the centre of mass and the shape of the support, not about how much of the biscuit is over the cup.

Step 1 — what is actually holding the biscuit up. The cup's rim is a circle, not a disc, so the biscuit touches it only where the rim passes underneath: at the two points where the rim circle crosses the biscuit's edge. Call them P and Q. Those two points are the whole support, and by the standard toppling argument the biscuit stays up while its centre of mass lies inside the convex hull of the contact points — here the region bounded by the chord PQ and the arc between them.



The geometry of the balance. The rim (radius R) crosses the biscuit's edge (radius b) at P and Q. The biscuit stays up while its centre of mass C lies inside the segment cut off by the chord PQ — so it falls the moment C reaches that chord.

Step 2 — write that condition as a distance. Let the cup's centre be O (rim radius $R = 3b/2$), the biscuit's centre of mass C (radius b), and $OC = d$. For two circles whose centres are d apart, the common chord — the line PQ — lies at distance

$$m = (d^2 + R^2 - b^2) / (2d)$$

from O. The biscuit's centre of mass is at distance d from O, so the limiting position is

$$m = d \Rightarrow d^2 + R^2 - b^2 = 2d^2 \Rightarrow d^2 = R^2 - b^2$$

Step 3 — evaluate with $R = 3b/2$.

$$d_{\min} = \sqrt{R^2 - b^2} = \sqrt{9b^2/4 - b^2} = \sqrt{5b^2/4} = (\sqrt{5}/2) b \approx 1.118 b$$

Step 4 — where does it start? “As far as possible from the cup's centre” while still balanced means the centre of mass is over the rim itself:

$$d_{\text{start}} = R = 1.5 b$$

(You can check the sense of that against Step 3: $d_{\text{start}} = 1.5b > d_{\min} = 1.118b$, so there is room to push.)

Step 5 — subtract.

$$\text{distance pushed} = d_{\text{start}} - d_{\min} = 1.5b - (\sqrt{5}/2) b = ((3 - \sqrt{5})/2) b \approx 0.382 b$$

Answer: D, $((3 - \sqrt{5})/2) b$.

Getting there without a calculator. The options are decimals, so at the very end you do have to turn a surd into one — but you never need long division. Recover $\sqrt{5}$ by squaring candidates: $2.2^2 = 4.84$ (too small), $2.3^2 = 5.29$ (too big), $2.24^2 = 5.018$ (just over), so $\sqrt{5} \approx 2.236$. Then

$$(3 - \sqrt{5})/2 \approx (3 - 2.236)/2 = 0.764/2 = 0.382$$

and 0.382 is option D. Notice what you did *not* need: 1.118b in Step 3, any decimal for $\sqrt{(5)}/2$, or a single division harder than halving 0.764.

The stronger check, which needs no number at all. Square to compare instead of evaluating. The answer must be $(3 - \sqrt{5})/2$, so it must satisfy $\sqrt{5} = 3 - 2 \times (\text{the answer})$. Test option A, 0.366: $3 - 2(0.366) = 2.268$, and $2.268^2 = 5.14 \neq 5$. Test option D, 0.382: $3 - 2(0.382) = 2.236$, and $2.236^2 = 5.00$ ✓. Squaring settles it exactly, and it is the technique to reach for whenever the options are decimals but the answer is a surd.

The limiting case that checks the whole derivation. Make the cup exactly as wide as the biscuit: $R = b$. Then the general result gives

$$d_{\min} = \sqrt{(R^2 - b^2)} = 0, \quad \text{so the push is } R - 0 = b$$

A biscuit the size of the rim can be pushed a full radius off centre before it topples — it is supported until its centre reaches the rim's centre. That is clearly right, and only an expression built on $\sqrt{(R^2 - b^2)}$ behaves that way.

Why the other options are wrong. All four are of the same shape — a surd over a small integer — so they cannot be separated by inspection. The two checks above do it: the answer must be smaller than $1.5b$, must vanish in the limit $b = R$, and evaluates at $0.382b$. Option A, $0.366b$, is close enough to tempt you and comes from using $\sqrt{3}$ in place of $\sqrt{5}$, i.e. from a wrong chord relation; B, $0.654b$, is much too large; C, $0.079b$, is much too small; E, $0.244b$, sits between but fails the $b = R$ test. Doing the general case first — $d_{\min} = \sqrt{(R^2 - b^2)}$ before substituting — is what makes this safe, because the general form can be checked against an easy case and the specific numbers cannot.

The habit. In any toppling question, ask first: what are the contact points, and what is the shape whose inside the centre of mass must lie in? A pencil on a table has one line of contact and falls at once; a box has an area and falls when its centre passes the edge; here the contact is an arc, and the answer is a chord condition. The shape of the support is the whole question.

Relevant topics: toppling and the centre of mass; stable equilibrium; the geometry of intersecting circles.

The trap. Treating the support as the overlap area between the biscuit and the cup. The rim is a circle, so the support is the arc where the rim passes under the biscuit, and balance fails when the centre of mass reaches the chord joining those two points.

Q15 · How many ice cubes the water can melt — **Answer B**

What is being tested. The two-stage thermal calculation — cool the water, melt the ice — and the discipline to keep an integer answer an integer.

Step 1 — how much energy the water can give up. The best case for melting ice is that the water is cooled all the way to 0°C :

$$Q_{\text{available}} = mc\Delta T = 0.5 \text{ kg} \times (4 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}) \times 20 \text{ K} \\ = 4.0 \times 10^4 \text{ J}$$

Step 2 — what one cube costs to melt. Ice at 0 °C needs no warming; it only needs melting:

$$Q_{\text{per cube}} = mL = 0.025 \text{ kg} \times (3 \times 10^5 \text{ J kg}^{-1}) = 7.5 \times 10^3 \text{ J}$$

Step 3 — divide, then round *down*.

$$n_{\text{max}} = (4.0 \times 10^4) / (7.5 \times 10^3) = 5.33\dots$$

The mixture must be **entirely liquid** at equilibrium, so there must be no ice left. Six cubes would need $6 \times 7.5 \times 10^3 = 4.5 \times 10^4 \text{ J}$, more than the water can supply, so some ice would survive at 0 °C and the mixture would be part ice. Five cubes is the largest number that can be fully melted.

Answer: B, 5.

Check that the fifth cube really does melt, and what is left over. Five cubes cost $5 \times 7.5 \times 10^3 = 3.75 \times 10^4 \text{ J}$, leaving

$$Q_{\text{left}} = 4.0 \times 10^4 - 3.75 \times 10^4 = 2.5 \times 10^3 \text{ J}$$

The final mixture is $0.500 + 5 \times 0.025 = 0.625 \text{ kg}$ of water, and $2.5 \times 10^3 \text{ J}$ warms it by

$$\Delta T = Q / (mc) = 2.5 \times 10^3 / (0.625 \times 4 \times 10^3) \approx 1 \text{ °C}$$

So the cup settles at about 1 °C, all liquid, with the whole calculation consistent. That is a useful thing to notice: the paper's rounded data (4×10^3 rather than 4200) makes the fifth-cube case land very close to 0 °C, which is exactly what "maximum" means.

Why the other options are wrong. C, 6, is the rounding error the question is built around: 5.33 rounds to 5 for a physical count, and rounding it up to 6 leaves ice floating in the cup — with the mixture then not entirely liquid, which is the condition the question set. D, 7, and E, 8, would leave several cubes of ice. A, 4, is what you get by making one of the two calculations slightly harder for yourself — for example by cooling the water only to some intermediate temperature, or by including a warming term for the melted ice when there is no energy left to do it.

The general shape of a melting-ice question. Energy available: $m_{\text{water}} c \Delta T$. Energy cost: $m_{\text{ice}} L$ (plus $m_{\text{ice}} c \Delta T$ afterwards, if the melt-water ends above 0 °C). The limiting number of cubes comes from dividing the first by the second and rounding towards zero whenever the question insists that nothing solid is left. Watch the units in one place: the mass must be in kilograms for both formulae, and 500 g and 25 g both invite you to forget that.

Relevant topics: specific heat capacity; latent heat of fusion; two-stage thermal equilibrium; integer constraints.

The trap. Rounding 5.33 up to 6. Six cubes need more energy than the water has, so ice would be left at 0 °C and the mixture would not be entirely liquid.

What is being tested. A three-line calculation where each line is a separate idea: refract the ray, slow it down, and measure the path — plus enough surd algebra to land on one of five similar-looking fractions.

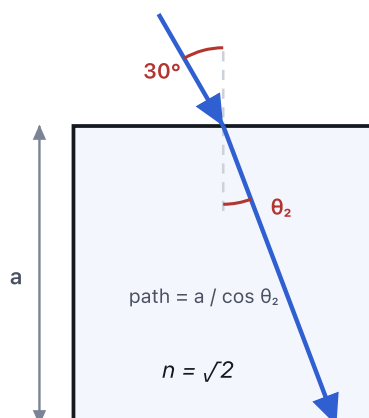
Step 1 — find the angle inside the glass. Here the 30° is the angle of incidence measured from the normal, as usual. Snell's law gives

$$\begin{aligned} n_1 \sin \theta_1 &= n_2 \sin \theta_2 \\ 1 \times \sin 30^\circ &= \sqrt{2} \times \sin \theta_2 \\ \sin \theta_2 &= 0.5 / \sqrt{2} = 1 / (2\sqrt{2}) \end{aligned}$$

Do not evaluate the angle. " $\theta_2 = 20.7^\circ$ " would need a calculator and nothing here needs it: keep $\sin \theta_2 = 1/(2\sqrt{2})$ and carry it forward. All you should conclude at this point is the direction — the ray bends *towards* the normal on entering the glass, because it enters a denser medium, so $\theta_2 < 30^\circ$. That single inequality is what does the work later.

Step 2 — how far it travels inside. The ray enters through one face and leaves through the opposite face, a distance a away. Because the ray is at angle θ_2 to the normal, the path is the hypotenuse of a right-angled triangle whose adjacent side is a :

$$\text{path} = a / \cos \theta_2$$



Why the path is $a / \cos \theta_2$. The ray enters the top face and leaves through the opposite face a distance a below it, so the distance it covers inside the glass is $a / \cos \theta_2$, at speed $c/\sqrt{2}$. The angle itself is left as θ_2 : this paper is non-calculator, and $\cos \theta_2$ is obtained exactly from $\sin^2 \theta_2 + \cos^2 \theta_2 = 1$ rather than from an evaluated angle.

Step 3 — the speed inside the glass.

$$v = c / n = c / \sqrt{2}$$

Step 4 — assemble, keeping $\cos \theta_2$ exact. Rather than evaluate 20.7° , use $\sin^2 + \cos^2 = 1$:

$$\cos \theta_2 = \sqrt{(1 - 1/8)} = \sqrt{(7/8)} = \sqrt{7} / (2\sqrt{2})$$

$$\begin{aligned} t &= \text{path} / v = [a / \cos \theta_2] \times [\sqrt{2} / c] \\ &= a \sqrt{2} / (c \cdot \sqrt{7}/(2\sqrt{2})) \\ &= 2\sqrt{2} \cdot \sqrt{2} \cdot a / (\sqrt{7} c) \\ &= 4a / (\sqrt{7} c) \end{aligned}$$

Answer: B, $(4/\sqrt{7}) \cdot a/c$. As a rough size, $\sqrt{7} \approx 2.65$ (because $2.65^2 = 7.02$), so the light takes about $4/2.65 \approx 1.5 \cdot a/c$ — longer than a/c , as it must be, since it travels further than a and it travels more slowly. But you can reach B without that division at all, which is the point of the next check.

Identifying B without evaluating anything. Two crude bounds, both from facts you already have.

Lower bound. The path through the glass is longer than a (the ray is not normal to the face), and the speed is $c/\sqrt{2}$, so $t > a/(c/\sqrt{2}) = \sqrt{2} \cdot a/c$.

Upper bound. You know $\theta_2 < 30^\circ$, so $\cos \theta_2 > \cos 30^\circ = \sqrt{3}/2$, so the path $a/\cos \theta_2 < 2a/\sqrt{3}$, and

$$t = (\text{path}) / (c/\sqrt{2}) < (2a/\sqrt{3}) \cdot (\sqrt{2}/c) = \sqrt{(8/3)} \cdot a/c$$

So the answer lies strictly between $\sqrt{2} = 1.414$ and $\sqrt{(8/3)} = 1.633$. Now compare the five options **by squaring them** — no roots, no decimals:

bounds squared:	2	$< t^2 < 8/3 = 2.67$
A $\sqrt{(8/7)} \rightarrow 8/7 = 1.14$		too small
B $4/\sqrt{7} \rightarrow 16/7 = 2.29$		✓ inside
C $\sqrt{2/3} \rightarrow 2/9 = 0.22$		too small
D $4/\sqrt{3} \rightarrow 16/3 = 5.33$		too big
E $4/\sqrt{2} \rightarrow 16/2 = 8.00$		too big

B is the only one that fits, and the decision used nothing but two inequalities and five easy divisions. This is what “non-calculator” actually asks of you: bound the answer, then test the options against the bounds. The full derivation above is how you *confirm* it; the bounds are how you *find* it in forty seconds.

Where the wrong options come from. A, $\sqrt{(8/7)} \cdot a/c = 1.069 \cdot a/c$, is exactly $a/(\cos \theta_2 \cdot c)$: the correct path, travelled at speed c — that is, with the refractive index forgotten in the speed but kept in the angle. It is the most instructive of the four wrong answers, because it is right in three places and wrong in one. C, D and E are the other combinations of the same ingredients: the depth a against the diagonal $a\sqrt{2}$, and the speed c against $c/\sqrt{2}$.

The method. Always write the time as $t = \text{path} / \text{speed}$ before substituting anything, and write both the path and the speed as multiples of a and c . Then the answer is forced to be $(\text{something}) \times a/c$, and the only question left is which surd the something is. That is exactly the form all five options take, which is a strong hint that the paper expects this route.

Relevant topics: Snell's law; speed of light in a medium; geometric path length; surd manipulation.

The trap. Using $v = c$ inside the glass, or taking the path as a rather than $a/\cos \theta_2$. Both errors give $\sqrt{(8/7)} \cdot a/c$, which is on the list.

Q17 · How many photoelectrons charge the capacitor — **Answer E**

What is being tested. Two ideas linked by one sentence: the capacitor stops collecting charge when its own voltage is high enough to stop the fastest photoelectrons, and the number of electrons is the charge divided by e — not the charge itself.

Step 1 — when does the flow stop? As electrons accumulate on the far plate, that plate goes negative relative to the illuminated one, and the potential difference opposes further electrons. The flow stops when the potential difference is exactly the stopping potential for the most energetic electrons:

$$eV_s = hf - hf_0 = h(f - f_0)$$

$$V_s = h(f - f_0) / e$$

Step 2 — the steady charge on the plates. When the p.d. across the capacitor reaches V_s ,

$$Q = C V_s = C h (f - f_0) / e$$

Step 3 — turn charge into a number of electrons. Each electron carries charge e , so

$$N = Q / e = C h (f - f_0) / e^2$$

Answer: E, $Ch(f - f_0)/e^2$.

Check the dimensions — five options, and only one survives. A capacitance has units of $C V^{-1}$, and since $1 V = 1 J C^{-1}$ we can write $F = C^2 J^{-1}$. Meanwhile hf is an energy, so $h(f - f_0)$ has units of J. Therefore

$$[C h(f - f_0)] = (C^2 J^{-1})(J) = C^2 \quad \Rightarrow [C h(f - f_0) / e^2] = \text{dimensionless} \checkmark$$

A number of electrons has to be dimensionless, and E is the only option that is. Option C comes out with units of $C^2/C = C$ — a charge, which is what option C in fact is: it is Q , the charge on the plates, one step short of the answer.

Where the wrong options come from. A and C add the two frequencies instead of subtracting — but the photoelectric effect is about the excess above the threshold, so a sum has no meaning here. B and D lose the Planck constant altogether: an answer containing no h cannot depend on the energy of the photon, and the whole physical content of the question is that it does. D also uses $f(f + f_0)$, mixing a frequency with a squared frequency against an e^2 — it fails the dimensional check twice over.

The trap worth naming. The difference between C and E is a single division by e , and it is the difference between “charge” and “number of electrons”. Read the question's last word. Whenever a question ends with “how many electrons”, “how many photons” or “how many particles”, expect the answer to require a division by the charge or the energy of one particle.

The wider point about this combination. Round 0 likes questions where two standard pieces of Year 12 physics meet — here the photoelectric effect and a capacitor. Neither part is hard; the mark is for noticing that the second part exists. Sketch the setup: light in, electrons out of one plate, onto the other, and the capacitor voltage grows until it forbids any more. That picture gives the sequence of the three steps above without any algebra.

Relevant topics: the photoelectric effect; stopping potential and the threshold frequency; $Q = CV$; dimensional checking.

The trap. Stopping at $Ch(f - f_0)/e$ — that is the charge on the plates. The question asks how many electrons, so divide by e once more.

What is being tested. Whether you reach for the small-angle approximations because the question offers $\alpha \ll 1$, or whether you notice that the answer is exactly zero and no approximation can improve on it.

Step 1 — write the range. For a projectile launched and landing at the same height,

$$R = v^2 \sin 2\theta / g$$

Step 2 — double both angles. The doubling is where the question is won:

$$2(\pi/4 + \alpha) = \pi/2 + 2\alpha$$

$$2(\pi/4 - \alpha) = \pi/2 - 2\alpha$$

Step 3 — use the complementary-angle identity.

$$\sin(\pi/2 + 2\alpha) = \cos 2\alpha$$

$$\sin(\pi/2 - 2\alpha) = \cos 2\alpha$$

Both ranges are $v^2 \cos 2\alpha / g$. Their difference is zero.

Answer: A, 0. And note the scope: this is not an approximation and it is not particular to small α . The two angles are complementary to 45° , and for any two angles that add to 90° the ranges are equal — that is one of the classic projectile results, and here it is being exploited with α small merely to make the pairs look less obviously related.

Why the answer is exactly zero — the calculus reading. The range is stationary at 45° :

$$dR/d\theta = (2v^2 / g) \cos 2\theta, \text{ which is zero at } \theta = 45^\circ$$

Because the top of the range-versus-angle curve is flat at 45° , moving the launch angle by $\pm\alpha$ changes the range only at order α^2 — and the two shifts give the same quadratic correction, so even that cancels. Nothing linear in α can appear.

That single observation kills four options at once. Options B, C, D and E are all proportional to α — first order in α . Since the range has a flat maximum at 45° , a first-order change is impossible, and all four must be wrong. On a question like this, the shape of the answer is enough; you do not need to do the algebra at all.

Where they come from, and why the paper put the approximations in the data sheet. The formula sheet offers $\sin \theta \approx \theta$, $\cos \theta \approx 1 - \theta^2/2$ and the binomial approximations. A tempting route is to expand $v^2 \sin 2\theta / g$ in α ; done carelessly — replacing $\cos 2\theta$ by 1 when differentiating, or keeping only one of the two shifts — it produces $2\alpha v^2/g$, $4\alpha v^2/g$ and their halves, which is exactly the wrong-option set. The approximations are on the sheet because other questions need them; the skill here is noticing that this one does not.

The habit this question rewards. When a formula contains a special angle — 45° , 30° , $\pi/4$ — look for the symmetry or the stationary point before you expand. One line of trigonometry beat a page of approximation, and the answer is exact rather than approximate.

Relevant topics: projectile range; the 45° maximum; stationary points; when not to use the small-angle approximations.

The trap. Expanding in α and picking a first-order term. The range is stationary at 45° , so the difference is exactly zero — options B, C, D and E are all linear in α and all impossible.

Q19 · Time for one third to decay — **Answer C**

What is being tested. Whether you read “one third decays” as “two thirds remain” — the single sentence that decides the question — and whether you can handle a logarithm with no calculator.

Step 1 — restate what is left.

$$\text{one third decayed} \Rightarrow 1 - 1/3 = 2/3 \text{ remains}$$

Step 2 — apply the decay law. After time t the fraction remaining is $(1/2)^{t/T}$:

$$(1/2)^{t/T} = 2/3$$

Step 3 — take logs of both sides.

$$\begin{aligned} (t/T) \log 2 &= \log(2/3) && \dots \text{taking logs and using } \log(1/2) = -\log 2 \\ t / T &= \log(3/2) / \log 2 \\ t &= T \log_2(3/2) \end{aligned}$$

Answer: C, $T \log_2(3/2)$.

Read the answer to check it — by bounding, not by calculating. You cannot evaluate $\log_2 3$ without a calculator, and you do not need to. Trap 3 between two powers of 2 you can compute in your head:

$$\begin{aligned} 2^{1.5} &= 2 \times \sqrt{2} = 2 \times 1.414 = 2.83 < 3 < 2^2 = 4 \\ \Rightarrow 1.5 &< \log_2 3 < 2 \\ \Rightarrow 0.5 &< \log_2 3 - 1 < 1 \\ \Rightarrow 0.5 &< \log_2(3/2) < 1, \quad \text{so } t \text{ is between } 0.5T \text{ and } 1T \end{aligned}$$

That is exactly what the physics demands: less than half the substance has decayed, so less than one half-life has elapsed — and more than none has, so more than zero. The bounds give you the answer's *whereabouts*, which is all a multiple-choice question ever asks for. (For the record $\log_2 3 = 1.585$ and $t = 0.585T$, but note that no step above needed either number.)

Why the other options are wrong. A, $(2/3)T$, assumes the decay is linear in time — at $t = (2/3)T$ it would have the substance decaying at a constant rate, which is not what radioactive decay does. D, $T \log_2 3$ (a little under $2T$, by the bound above), is the answer to a different question: it is the time for two thirds to have decayed, i.e. one third remaining, because it comes from $(1/2)^{t/T} = 1/3$. That is the single most common slip here, and it is why the question says “one third decay” rather than “one third remaining”. E, $T^{2/3}$, is dimensionally impossible — a half-life raised to a power is not a time — so it dies on inspection. B, $\sqrt{(2/3)} T \approx 0.82T$, is a third plausible-looking number with no derivation behind it; you cannot get a square root out of an exponential decay law.

Carry the general form. If a fraction f remains,

$$t = T \log_2(1/f)$$

Check it on two cases you know by heart: $f = 1/2$ gives $t = T \log_2 2 = T$ ✓, and $f = 1/4$ gives $2T$ ✓. A general formula that reproduces the standard cases is worth remembering, and these two are the ones it will be tested against.

How to do $\log_2(3/2)$ without a calculator. Split it: $\log_2(3/2) = \log_2 3 - 1$, and $\log_2 3$ is between $\log_2 2 = 1$ and $\log_2 4 = 2$, closer to 1.5 than to 2 — so for the record 0.585 is plausible before you compute anything. If the options had required the value rather than the expression, that bracket would be enough to choose.

Relevant topics: exponential decay; half-life; the decay law in index form; logarithms.

The trap. Reading one third decayed as one third remaining and answering $T \log_2 3 \approx 1.6T$. Two thirds remains, so $t = T \log_2(3/2)$, which the bounds put between $0.5T$ and $1T$ — less than a half-life is wrong; under one half-life is right.

Q20 · Jerk of a bungee jumper against time — Answer D

What is being tested. The shape of a graph you have never seen before, built from two pieces of ordinary physics: free fall, then a spring. Every wrong option breaks one specific feature of that shape, so the method is to establish the features first.

Phase 1 — the rope is slack. The jumper is in free fall, so $a = g$, a constant. Therefore

$$j = da/dt = 0 \quad \text{all the way through phase 1}$$

The moment the rope goes taut. This is the key instant, and the dashed line in options A, D and E marks it. At that instant the extension is zero, so the spring force is zero, and the acceleration is still g — the acceleration itself is continuous. But from that instant the acceleration begins to change, so the jerk jumps **discontinuously** from 0 to a finite value.

That one sentence removes options B and C. Both of them leave the origin of jerk at zero at the moment the rope goes taut, which would mean the jumper's acceleration carries on unchanged even though a spring has just started pulling on it. And C is piecewise straight, which would require the acceleration to change quadratically with time — a linear jerk is not what a spring produces.

Phase 2 — the rope stretches. Take downwards as positive and let x be the extension:

$$\begin{aligned} ma &= mg - kx & \Rightarrow & & a &= g - \omega^2 x, & \omega^2 &= k/m \\ j &= da/dt = -\omega^2 v \end{aligned}$$

So the jerk is proportional to the jumper's speed (with a minus sign, i.e. it acts to reduce the acceleration).

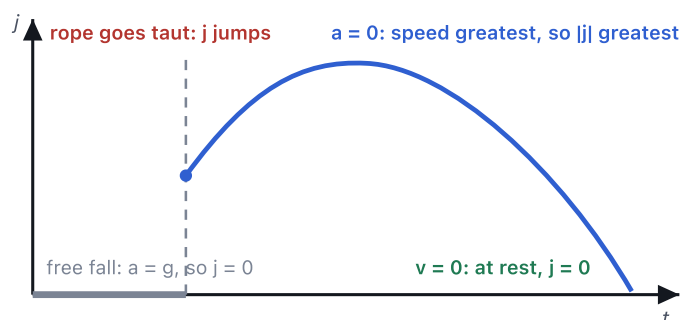
What does the speed do just after the rope goes taut? At that instant the acceleration is g , still downwards, so the jumper is *still speeding up*. The speed keeps growing for a moment. Therefore $|j| \propto v$ grows too: the curve must **rise** before it falls. This is the feature that separates D from E.

How the phase ends. The jumper's speed peaks when the acceleration passes through zero — at the equilibrium position of the oscillation that follows — and then decreases. The phase ends at the lowest point, where the jumper is instantaneously at rest:

$$v = 0 \Rightarrow j = -\omega^2 v = 0$$

So the curve must come back to exactly zero at the end, and it never changes sign in between, because the velocity is always downwards.

Answer: D — the graph that jumps to a finite value at the dashed line, rises to a rounded maximum, and returns smoothly to zero.



The graph the correct option has to have. Flat at zero while the rope is slack; a discontinuous jump at the instant it goes taut; a rise to a maximum as the jumper is still speeding up; and a return to exactly zero at the moment of instantaneous rest.

Why E is wrong even though it has the jump. E starts at a high value at the dashed line and decays steadily towards the axis. It is missing the initial rise — and, more fundamentally, a monotonic decay of that kind is the signature of an exponential, which comes from a first-order process. The motion after the rope goes taut is simple harmonic, whose speed is a sinusoid, so the jerk must rise to a maximum and return to zero at a definite time, not creep towards the axis.

Why A is wrong. A jumps and then falls to zero in a straight line, so the jerk would be linear in time. Since $j \propto v$, that would force the speed to be linear in time — constant acceleration — which is the opposite of what a spring does.

The method for any “which graph” question. Do not try to plot the function. Instead, list the features the correct graph must have and test each option against them:

- | | |
|---|-----------------|
| 1. flat at zero during free fall | (all five pass) |
| 2. a jump, not a smooth start, at the rope's tautness | (removes B, C) |
| 3. an initial rise before the fall | (removes E, A) |
| 4. a return to exactly zero at instantaneous rest | (removes E) |

Four features, and each option fails at a named one. That is worth more than any amount of sketching, and it is how this question is meant to be answered in the two and a half minutes available.

Relevant topics: Newton's second law with a spring; Hooke's law; the onset of simple harmonic motion; graph-shape reasoning; rates of change.

The trap. Assuming the jerk is zero when the rope goes taut because the acceleration is momentarily unchanged. The acceleration is continuous, but its *rate of change* jumps — that is the whole point of the question.

Q21 · Moments — **Answer A**

What is being tested. One realisation unlocks the whole question: **the rod does not move.** "The hinge fixes O and the inextensible vertical string fixes the upper end, so a rigid rod of fixed "length is held at 30° for ever. The ring is therefore just a bead sliding down a smooth incline "that happens to be held up by a string — and the tension is found from moments, not from forces "on the ring.

Step 0 — read the given form before doing any algebra. You are told $T = mg(1 - kt^2)$. "At $t = 0$ the ring is at the upper end, i.e. a distance ℓ from the hinge, "and the given form says $T = mg$. So whatever you derive must collapse to mg "when the ring sits at the top. That single check is worth doing first, because it fixes the "normalisation and it kills several candidate mistakes immediately.

Step 1 — how fast does the ring move? The rod is smooth, so the only force with a component along "it is gravity:

$$a = g \sin 30^\circ = g / 2$$

Released from rest, the distance s slid down from the upper end is

$$s = \frac{1}{2} a t^2 = \frac{1}{2} (g/2) t^2 = g t^2 / 4$$

so its distance from the hinge O is

$$r = \ell - s = \ell - g t^2 / 4$$

Step 2 — the normal reaction. The ring has no acceleration perpendicular to the rod, so the "perpendicular component of gravity is balanced by the normal reaction:

$$N = mg \cos 30^\circ$$

Step 3 — moments about O for the rod. The rod is weightless and static, so the moments must balance. "Two forces act at a distance from O:

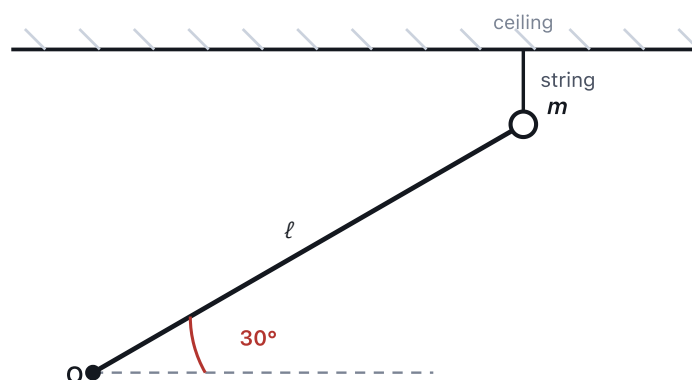
tension T , vertical, at the upper end $\rightarrow$ lever arm $\ell \cos 30^\circ$
 normal reaction N , perpendicular to the rod, at distance $r \rightarrow$ lever arm r

$$T \cdot \ell \cos 30^\circ = N \cdot r = mg \cos 30^\circ \cdot r$$

and the $\cos 30^\circ$ cancels — which is the pretty part of this question, because it means the "30° never survives into the answer:

$$T = mg r / \ell = mg (\ell - g t^2/4) / \ell = mg (1 - g t^2 / 4\ell)$$

Answer: A, $k = g / 4\ell$.



The rod is held still by the string; the ring slides along it. Note which end the ring starts at, and that the rod keeps the ring at a fixed angle — the motion is one-dimensional along the rod, not vertical.

The $t = 0$ check works. Putting $r = \ell$ gives $T = mg$, exactly as "the given form requires. And as the ring reaches the hinge, $r \rightarrow 0$ and the tension "falls to zero — with no ring left on the rod there is nothing for the string to hold up.

Dimensions, as a second check. kt^2 must be dimensionless, so k is a " $1/\text{time}^2$ ". Every option is g divided by a length, which is exactly " s^{-2} ", so dimensions alone cannot separate them — you have to get the factor right. "That is deliberate.

Where the wrong options come from. All five options are g/ℓ times a pure number, so "every wrong answer is one specific slip in Step 1:

A	$0.25\ g/\ell$	correct:	$s = \frac{1}{2}(g \sin 30^\circ)t^2 = g\ t^2/4$
B	$0.50\ g/\ell$	took $a = g$:	$s = \frac{1}{2} g\ t^2$ (forgot $\sin 30^\circ$)
C	$1.00\ g/\ell$	took $a = g$, forgot $\frac{1}{2}$:	$s = g\ t^2$ (two slips)
D	$0.71\ g/\ell$	a spurious $\sqrt{2}$	
E	$0.87\ g/\ell$	a spurious $\sqrt{3}$ — e.g. reaching for $\cos 30^\circ$ instead of $\sin 30^\circ$	

B is by far the most tempting, because "released from rest, so $s = \frac{1}{2}gt^2$ " is an reflex. It is "wrong here only because gravity does not act along the rod.

A third route, if you distrust the moment arm. Resolve instead of taking moments about O: the rod is "weightless and static, so the hinge reaction plus T plus N must sum to "zero. Taking moments is faster and it removes the hinge reaction entirely, which is why it is the "right first move on any question involving a hinged body.

Relevant topics: moments about a hinge; resolution along and perpendicular to an incline; " $s = \frac{1}{2}at^2$ " from rest; reading a given algebraic form for a boundary condition.

The trap. Taking the rod to rotate, or treating the ring as falling vertically with $a = g$. The rod is held still by the string; the ring accelerates at $g \sin 30^\circ = g/2$ along it.

Q22 · Conical pendulum — Answer A

What is being tested. Whether you know the conical-pendulum result well enough to use it " "as a ratio, and whether you remember that it gives ω^2 , not ω . The "question is deliberately phrased as "by what factor", which is a loud hint that nothing except "the ratio matters — ℓ , g and m all cancel.

Step 1 — the standard result, derived in two lines. With the string at angle θ to the "vertical" and the bob on a circle of radius $r = \ell \sin \theta$:

vertical: $T \cos \theta = mg$

horizontal: $T \sin \theta = m \omega^2 r = m \omega^2 \ell \sin \theta$

The second line gives $T = m \omega^2 \ell$ directly (the $\sin \theta$ cancels on both sides). "Substituting into the first:

$$m \omega^2 \ell \cos \theta = mg \quad \Rightarrow \quad \omega^2 = g / (\ell \cos \theta)$$

Step 2 — take the ratio, not the values. For two angles at the same ℓ :

$$\omega_2 / \omega_1 = \sqrt{[\cos \theta_1 / \cos \theta_2]} = \sqrt{[\cos 30^\circ / \cos 60^\circ]}$$

$$\cos 30^\circ = \sqrt{3} / 2, \quad \cos 60^\circ = 1 / 2$$

$$\cos 30^\circ / \cos 60^\circ = \sqrt{3}$$

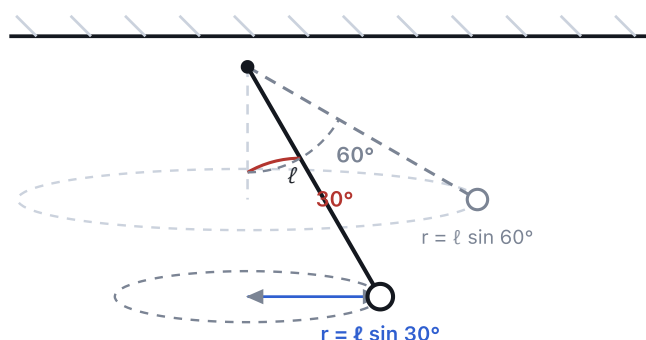
$$\omega_2 / \omega_1 = \sqrt{\sqrt{3}} = 3^{1/4} \approx 1.32$$

Answer: A, $\sqrt{\sqrt{3}}$.

You can pick it out without evaluating the fourth root at all. Raise each option to the fourth power — the answer must give exactly 3, and fourth powers are just squaring twice:

A	$\sqrt{\sqrt{3}}$	$\rightarrow$	$(\sqrt{\sqrt{3}})^4 = 3$	✓
B	$\sqrt{2}$	$\rightarrow$	$(\sqrt{2})^4 = 4$	✗
C	$\sqrt{3}$	$\rightarrow$	$(\sqrt{3})^4 = 9$	✗
D	2	$\rightarrow$	$2^4 = 16$	✗
E	3	$\rightarrow$	$3^4 = 81$	✗

One line, no arithmetic beyond $2^2 = 4$ and $3^2 = 9$. If you do want the size of it, take it in two easy square roots: $\sqrt{3} = 1.732$, then find the number whose square is 1.732 — $1.3^2 = 1.69$ is low and $1.32^2 = 1.742$ is high, so it is about 1.316. Either way you land on A.



Doubling the angle from 30° to 60° nearly doubles the radius — $\ell \sin 30^\circ = 0.50\ell$ becomes $\ell \sin 60^\circ = 0.87\ell$ — while the height of the bob above the pivot falls from 0.87ℓ to 0.50ℓ . Every one of those changes is packed into $\omega^2 = g / (\ell \cos \theta)$.

Why the answer is so small — the physical check. Doubling the angle from 30° to 60° sounds like a "dramatic change, and two things do change a lot: the radius grows from 0.50ℓ to 0.87ℓ and the bob

drops from 0.87ℓ to 0.50ℓ below the "pivot. But $\omega^2 \propto 1/\cos \theta$, and $\cos \theta$ only falls by a factor of $\sqrt{3} \approx 1.73$. Taking the square root halves that in log terms and leaves a factor of "just 1.32. If you find yourself writing down "about 1.7" or "about 2", you have "most likely reported ω^2 's ratio as if it were ω 's.

Where the wrong options come from.

$C = \sqrt{3} = 1.732$	the exact ratio of ω^2 – the square root was forgotten
$E = 3$	$\tan 60^\circ / \tan 30^\circ$ – reaching for tan instead of cos
$D = 2$	the angle itself doubled, so "the factor must be 2"
$B = \sqrt{2} = 1.414$	a near-miss decoy sitting just above the true 1.32

C is the one to worry about: it is a fully correct calculation of the wrong quantity. Whenever a question "asks "by what factor" about a quantity that appears squared, write down explicitly whether you "have just found X or X^2 before you look at the options.

The method. For any " θ changes, what happens to ω " question, rearrange "the governing equation so the changing quantity stands alone:

$$\omega = \sqrt{(g / (\ell \cos \theta))} \propto (\cos \theta)^{-1/2}$$

Then the factor is $(\cos \theta_1 / \cos \theta_2)^{1/2}$ and no other quantity can appear. That is why the "options are pure numbers with no g or ℓ in them — a useful clue that "you are on the right track.

Relevant topics: conical pendulum; resolving tension into vertical and radial components; " $\omega^2 = g/(\ell \cos \theta)$ "; ratio reasoning with a square root.

The trap. Reporting $\cos 30^\circ / \cos 60^\circ = \sqrt{3}$ as the answer. That is the factor for ω^2 ; the question asks about ω , so you must take the square root: $\sqrt{\sqrt{3}}$.

Q23 · Log-log graphs — Answer B

What is being tested. Two separations that are usually taught apart and are here "combined: reading a power law off a log-log plot, and knowing how a point source's intensity "falls with distance. Neither step is hard; the difficulty is that the answer is not a round "number, so you cannot guess it — you have to commit to the arithmetic.

Step 1 — read the gradient off the log-log plot. On log axes a power law $R \propto I^n$ is a "straight line of gradient n . The line runs from $(0, 5.2)$ to $(4, 1.8)$, so

$$n = (1.8 - 5.2) / (4 - 0) = -3.4 / 4 = -0.85$$

$$R \propto I^{-0.85}$$

Note the two things that make log axes worth using: the intercept (the "5.2") is irrelevant to this "question, and the units inside the logarithm are handled for you by the axis labels.

Step 2 — what halving the distance does to the intensity. A point source spreads its power over a "sphere of area $4\pi d^2$, so

$$I \propto 1 / d^2 \quad \Rightarrow \quad d \rightarrow d/2 \text{ gives } I \rightarrow 4I$$

Step 3 — combine.

$$R_{\text{new}} / R_0 = (4I / I)^{-0.85} = 4^{-0.85}$$

Now bracket it — do not try to compute it. $4^{-0.85}$ has no mental evaluation, and it does not need one. The exponent -0.85 lies between -1 and $-3/4$, and for a base bigger than 1 a more negative power is a smaller number, so

$$\begin{array}{ccccc} 4^{-1} & < & 4^{-0.85} & < & 4^{-3/4} \\ 1/4 & < & 4^{-0.85} & < & 1 / 4^{3/4} \end{array}$$

and the right-hand bound is easy: $4^{3/4} = (2^2)^{3/4} = 2^{3/2} = 2\sqrt{2} = 2 \times 1.414 = 2.83$, so

$$0.25 < 4^{-0.85} < 1/2.83 = 0.354$$

Of the five options — 0.1, 0.3, 0.5, 0.7, 0.9 — **only 0.3 lies between 0.25 and 0.354**. The question is settled by two powers you can do in your head, and the gradient never had to be read to better than "about -0.9 ". That is the whole trick of this paper: bound the answer, then let the options do the rounding.

Answer: B, 0.3 R_0 .

The robustness check that matters most here. Suppose you read the gradient as exactly -1 instead of -0.85 — a very reasonable mis-reading of a hand-drawn "graph. Then

$$R_{\text{new}} / R_0 = 4^{-1} = 0.25$$

and the nearest option is still $0.3 R_0$. So the answer does not depend on reading the gradient "to two decimal places; it only depends on the gradient being close to -1 and clearly "not close to 0 or -2 ". That is what "a good approximation" in the "question is telling you.

Where the wrong options come from.

C = 0.5 R_0	used $I \propto 1/d$, not $1/d^2 \rightarrow 2^{-0.85} \approx 0.55$
A = 0.1 R_0	used $I \propto 1/d^3$, or $4^{-1.7} \approx 0.1$
D = 0.7 R_0	treated the change as small — as if the axes were linear
E = 0.9 R_0	almost no change at all

C is the one to watch for: "halve the distance, so halve the intensity" is the single most common error "in this topic, and the option is sitting there to reward it.

The method for any log–log question. Do three things in this order, and never multiply numbers "before you have done all three:

1. gradient of the line $\rightarrow$ the power n in $y \propto x^n$
2. what the physics does to $x \rightarrow$ the factor by which x changes
3. raise that factor to the power n

Relevant topics: log–log plots and power laws; inverse-square law for a point source; estimating "powers of 2 without a calculator.

The trap. Halving the distance and halving the intensity — that error leads straight to option C, $0.5R_0$ — or trying to evaluate $4^{-0.85}$ directly. Bracket it: it lies between $4^{-1} = 0.25$ and $4^{-3/4} = 1/(2\sqrt{2}) = 0.354$, and 0.3 is the only option in that window.

Q24 · Centre of mass — Answer A

What is being tested. Whether you see that the ladder's gravitational potential energy "depends on exactly one thing — the height of its centre of mass — and whether you can get that "height change from the geometry without differentiating anything. The word "small" is the key: "it tells you to use a first-order (linear) approximation, so a single derivative is the whole "calculation.

Step 1 — the only variable that matters. The ladder is uniform, so its centre of mass is at its "midpoint. If the ladder has length L and makes angle θ with the "horizontal, the centre of mass is at height

$$h = (L / 2) \sin \theta \quad \Rightarrow \quad U = mg (L/2) \sin \theta$$

and the foot of the ladder is a horizontal distance $x = L \cos \theta$ from the wall.

Step 2 — relate a change in x to a change in h . Differentiate both with "respect to θ and divide:

$$\begin{aligned} dh/d\theta &= (L/2) \cos \theta, & dx/d\theta &= -L \sin \theta \\ dh/dx &= -(1/2) \cot \theta \end{aligned}$$

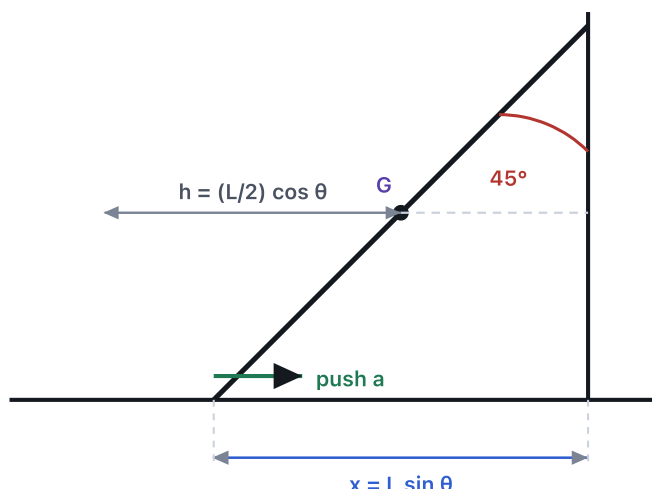
The minus sign is the physics: pushing the foot *towards* the wall (x decreasing) makes "the centre of mass go *up*.

Step 3 — evaluate at 45° . Here "45° to the vertical" is the same thing as 45° to the horizontal, "since the wall and the floor are perpendicular. So $\cot 45^\circ = 1$:

$$\Delta h = -(1/2) \cot 45^\circ \cdot \Delta x = -(1/2)(-a) = a / 2$$

$$\Delta U = mg \Delta h = (1/2) m g a$$

Answer: A, $\frac{1}{2} mga$.



The centre of mass is half way up the ladder, so its height is $h = (L/2) \cos \theta$. At 45° a small push a towards the wall raises it by $(a/2) \tan 45^\circ = a/2$ — the top of the ladder rises by the full a , the centre by half of that.

The tidy check. The centre of mass rises by exactly half of a — half because the centre of mass is at the middle of the ladder. Here is the same result without calculus: when the foot moves in by a , the top of the ladder slides up the wall by a at 45° (the top and foot swap equal amounts of horizontal and vertical). The middle rises half as much as the top, so $\Delta h = a/2$. Two routes, same answer.

The general result, and why it makes sense.

$$\Delta U = \frac{1}{2} m g a \cot \theta$$

This is the sanity check that catches most slips. If the ladder is nearly vertical ($\theta \rightarrow 90^\circ$), $\cot \theta \rightarrow 0$: nudging the foot barely lifts the middle at all, which is right. If the ladder is nearly flat ($\theta \rightarrow 0$), $\cot \theta \rightarrow \infty$: a tiny push lifts the middle enormously, which is also right. At 45° the factor is exactly 1, so the answer is the clean-looking $\frac{1}{2}mga$. Any answer with a $\sqrt{2}$ in it should make you suspicious, because 45° is precisely the angle at which the sine and cosine are equal and the square roots cancel.

Where the wrong options come from.

$C = mga$	forgot the centre of mass is at the middle (used the top of the ladder)
$B = mga/\sqrt{2}$	picked up a spurious $\sin 45^\circ$ factor
$D = \sqrt{2} mga$	picked up a spurious $1/\cos 45^\circ$ factor
$E = 2 mga$	over-corrected in both directions at once

C is the trap with real physics content: it is the change in potential energy of the *top* of the ladder, not of the ladder. Whenever a uniform body is involved, the factor of $\frac{1}{2}$ from the centre of mass is the single easiest thing to lose.

Relevant topics: centre of mass of a uniform body; gravitational potential energy mgh ; small-change approximations; the geometry of a ladder against a wall.

The trap. Forgetting that only the centre of mass matters, which gives mga (option C). Also note that 45° to the vertical is 45° to the horizontal, so $\cot \theta = 1$.

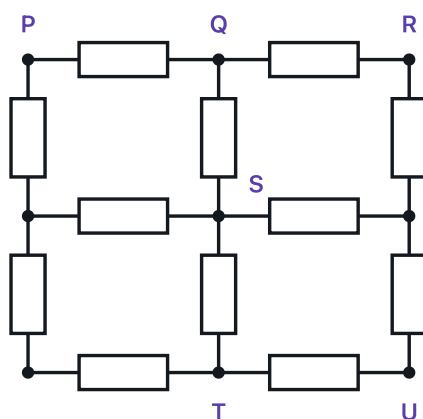
Q25 · Network reduction — Answer E

What is being tested. This looks like five brutal network calculations, and it is "deliberately built so that it is not. Every one of the five pairs sits on a symmetry axis of the "grid, so each can be reduced by **folding** — and once folded, three of the five collapse to "plain series and parallel. Recognising the symmetry is the entire question.

Set-up. Call every resistor $1\ \Omega$; only the ratios matter. Label the nine nodes as a 3×3 grid:

P — Q — R	row 0
• — S — •	row 1
• — T — U	row 2

Twelve resistors: six horizontal (two per row) and six vertical (two per column).



A 3×3 grid of nodes with all twelve resistors identical. The labels sit on the nodes: P, Q, R along the top; S at the centre; T and U along the bottom. Every line you can see is one resistor.

The two symmetries.

M : reflect in the vertical centre line	$P \leftrightarrow R$, $A \leftrightarrow B$, $C \leftrightarrow U$, and Q, S, T fixed
D : reflect in the diagonal P–S–U	$Q \leftrightarrow A$, $R \leftrightarrow C$, $B \leftrightarrow T$, and P, S, U fixed

Under **M** a mirrored pair of resistors becomes two in parallel, i.e. $\frac{1}{2}\ \Omega$. Under **D** "the same is true. That single rule does all the work.

QT — reduce by M. Q and T are both fixed by M, so the excitation is symmetric and " $V(P) = V(R)$, $V(A) = V(B)$, $V(C) = V(U)$ ". Fold:

Q–P and Q–R $\rightarrow \frac{1}{2}\ \Omega$	Q–S stays $1\ \Omega$
A–S and S–B $\rightarrow \frac{1}{2}\ \Omega$	S–T stays $1\ \Omega$
C–T and T–U $\rightarrow \frac{1}{2}\ \Omega$	
P–A and R–B $\rightarrow \frac{1}{2}\ \Omega$	A–C and B–U $\rightarrow \frac{1}{2}\ \Omega$

The folded circuit is a Wheatstone bridge whose upper arms are $\frac{1}{2} + \frac{1}{2} = 1\ \Omega$ (via P, A) and " $1\ \Omega$ (via S), and whose lower arms are $1\ \Omega$ (via C) and $1\ \Omega$ — so it is **balanced** and the $\frac{1}{2}\ \Omega$ link between A and S carries no current. "Remove it:

$$R_{QT} = (1 + 1) \parallel (1 + 1) = 2 \parallel 2 = 1 \, \Omega$$

QS — reduce by M. Same fold; now measure between Q and S. The direct resistor "Q–S = 1 Ω " is in parallel with everything else, so the answer must be *below* "1 Ω ". Solving the same folded bridge (inject 1 A at Q, extract at S) gives

$$\begin{aligned} V_C &= \frac{3}{4} V_A, & V_A &= (V_Q + 2 V_S)/4, & V_S &= (V_Q + 2 V_A)/4 \\ \Rightarrow V_A &= 2V_Q/7, & \text{and at Q: } &2(V_Q - 9V_Q/14) + V_Q &= 1 \\ \Rightarrow V_Q &= 7/12 \end{aligned}$$

$$R_{QS} = 7/12 \, \Omega$$

PR — reduce by M, antisymmetrically. Now P and R are a mirrored *pair*, so the excitation is "antisymmetric: $V(Q) = V(S) = V(T) = \frac{1}{2}(V_P + V_R)$ ", and those three axis nodes sit at "the same potential, so the links Q–S and S–T carry no current. Consider the left half with the "axis held at $V/2$ ", taking $V_P = V$:

$$\begin{aligned} P-A &= 1, & A-C &= 1, & P\text{-axis} &= 1, & A\text{-axis} &= 1, & C\text{-axis} &= 1 \\ \text{node A: } &3V_A - V_C &= 3V/2 \\ \text{node C: } &2V_C &= V_A + V/2 &\Rightarrow V_A &= 7V/10, & V_C &= 3V/5 \\ I \text{ from P} &= (V - V_A) + (V - V/2) = 0.3V + 0.5V = 4V/5 \end{aligned}$$

$$R_{PR} = V / (4V/5) = 5/4 \, \Omega = 1.25 \, \Omega$$

PS and PU — reduce by D. Both P and U are fixed by the diagonal reflection, so fold along the "diagonal: Q merges with A, R with C, "B with T, and every merged pair of resistors becomes $\frac{1}{2} \, \Omega$ ". The folded circuit is a chain with one bypass:

$$\begin{array}{ccccccc} P & \xrightarrow{\frac{1}{2}} & QA & \xrightarrow{\frac{1}{2}} & RC & \xrightarrow{\frac{1}{2}} & TB & \xrightarrow{\frac{1}{2}} & U \\ & & QA & \xrightarrow{\frac{1}{2}} & S & \xrightarrow{\frac{1}{2}} & TB & & \end{array}$$

For **PU** (measure P to U): by symmetry $V_S = V_{RC}$, and solving gives " $V_{RC} = \frac{1}{2}V_P$, $V_{QA} = \frac{2}{3}V_P$ ", so with 1 A injected, " $2(V_P - \frac{2}{3}V_P) = 1$ " and

$$R_{PU} = 3/2 \, \Omega = 1.5 \, \Omega$$

For **PS** (measure P to S): the same folded circuit gives $V_{QA} = 3/8$, " $V_{RC} = 1/4$, $V_{TB} = 1/8$ " with 1 A injected at P, so

$$R_{PS} = 7/8 \, \Omega = 0.875 \, \Omega$$

Now take the median.

$$QS = 7/12 \quad PS = 7/8 \quad QT = 1 \quad PR = 5/4 \quad PU = 3/2$$

Order them **by cross-multiplying, never by converting to decimals** — that is the whole point of keeping them as fractions. $7/12$ against $7/8$: compare $7 \times 8 = 56$ with $7 \times 12 = 84$, so $7/12 < 7/8$. $7/8$ against 1 : $7 < 8$. 1 against $5/4$: $4 < 5$. And $5/4$ against $3/2$: $10 < 12$. So the order is

$$\begin{array}{ccccccccc}
 7/12 & < & 7/8 & < & 1 & < & 5/4 & < & 3/2 \\
 \text{QS} & & \text{PS} & & \text{QT} & & \text{PR} & & \text{PU} \\
 & & & & \uparrow & & & & \\
 & & & & \text{the median} & & & &
 \end{array}$$

Answer: E, QT.

Two shortcuts that get you there faster. First, you never needed PS, PU or PR to any precision: "QS is obviously the smallest (Q and S are joined by a single resistor with the rest of the grid in "parallel, so it is under $1\ \Omega$), and PU is obviously the largest (P and U are opposite "corners, the longest journey through the grid). PR and PS bracket $1\ \Omega$. So the median "is whichever of PS, QT, PR is the middle one — and since $R_{PS} < 1$ "(P and S are diagonal neighbours, closer than the opposite corners) while R_{PR} spans "a whole row, QT at exactly $1\ \Omega$ is the only candidate that can sit in the middle.

Second, and worth internalising: **QT = $1\ \Omega$ exactly**, i.e. the whole grid measured across its axis is "equivalent to a single resistor. That is not a coincidence — it is the balanced-bridge result, and "it is the sort of clean number a paper setter builds a question around.

Where the wrong options come from. Each names a different pair, so there is no arithmetic to slip "on; the only way to get this wrong is to guess the ordering instead of establishing it. The "ordering is by "how far apart are the two nodes in the grid", tempered by how many parallel "routes exist — which is why QT, a two-step journey straight down the axis with a balanced bridge, "lands exactly in the middle.

Relevant topics: symmetry folding of resistor networks; series and parallel combinations; the "balanced Wheatstone bridge; identifying a median without evaluating every term.

The trap. Trying to evaluate all five networks by brute force, or assuming the median follows the alphabetical order of the labels. Fold on the symmetry axis first — QT reduces to a balanced bridge and is exactly $1\ \Omega$.